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Abstract

The accurate identification of protein-ligand docking sites is a central challenge in modern computational drug discovery and structure-based molecular design. Due to the combinatorially large search space presented by complex target proteins, classical algorithms often face severe performance bottlenecks. This report presents Q-MODE (Quantum Molecular Design), a multidisciplinary software pipeline that bridges advanced classical chemoinformatics with quantum algorithms. The system is conceptually divided into two major components. First, an automated classical preprocessing module extracts three-dimensional interaction features from experimentally derived Protein Data Bank (PDB) files, clusters these chemical properties, and optimally maps them onto a discrete two-dimensional interaction lattice while preserving covalent molecular topology. Second, a quantum computing module, implemented via IBM Qiskit, translates these mathematical lattices into qubit registers using dual encoding schemes. A modified Grover’s search algorithm rapidly isolates candidate docking sites from a uniform quantum superposition of all possible protein sub-segments. Finally, a continuous amplitude encoding enables a quantum SWAP test to calculate the precise Euclidean distance in Hilbert space, accurately ranking the binding affinities of the candidate sites. By synthesizing chemical graph theory, linear assignment optimization, and quantum search dynamics, Q-MODE provides a scalable, end-to-end framework for near-term quantum chemoinformatics.

1. Introduction

Protein-ligand docking involves predicting the preferred orientation of a ligand molecule to a target protein when bound to each other to form a stable complex. Knowledge of the preferred orientation may be used to predict the strength

of association or binding affinity between the two molecules, which is critical for rational drug design. However, mapping the vast surface area and conformational flexibility of proteins against varied ligand geometries results in an exponentially scaling computational problem. The classical exploration of this search space typically relies on heuristic approximations such as simulated annealing or genetic algorithms, which cannot guarantee the discovery of the global energy minimum.

To overcome the limitations of classical heuristic searches, quantum computing algorithms offer profound theoretical advantages, notably through amplitude amplification and quantum phase estimation. This project builds upon the theoretical framework introduced by Liliopoulos et al., which proposes mapping the docking interaction space onto a discrete grid and utilizing Grover’s algorithm to search for specific interaction patterns.

However, a critical gap exists in the literature: theoretical quantum models assume the existence of an "inner lattice" where continuous, high-dimensional biochemical features are already discretized and perfectly aligned. The translation from real-world 3D atomic coordinates (e.g., from X-ray crystallography or Cryo-EM) to this idealized 2D quantum-ready lattice is non-trivial.

The Q-MODE multidisciplinary project directly addresses this gap by implementing a robust classical-to-quantum pipeline. It integrates chemoinformatics libraries (RDKit), spatial clustering, graph traversal algorithms, dimensionality reduction, and quantum simulation (Qiskit). This report comprehensively details the architectural decisions, algorithmic implementations, and the underlying mathematical models of both the classical amino acid lattice mapping and the quantum interaction site matching.

2. Classical Chemoinformatics Preprocessing

The classical module, referred to as the Amino Acid Lattice Mapping pipeline, is responsible for digesting raw structural data and producing the mathematically rigorous, discrete sequences required by the quantum search registers. This involves a multi-stage process from feature extraction to collision-free grid snapping.

2.1. PDB Parsing and Molecular Instantiation

The pipeline is initialized by reading target protein or isolated binding pocket data from .pdb files. Since standard parsing often strips critical bonding information, we employ a hybrid instantiation strategy. For standard amino acids, we fallback to canonical SMILES representations to generate an accurate initial molecular topology graph.

Using the RDKit toolkit, we generate a 3D conformer using the ETKDGV3 (Experimental-Torsion-Knowledge Distance Geometry) algorithm. To retain the actual experimental geometry of the pocket, we perform a deterministic overlay: the theoretical coordinates of the generated conformer’s heavy atoms are completely overwritten by the real coordinates parsed from the PDB file.

Listing 1: Deterministic overlay of PDB coordinates onto RDKit conformer

```
def _overlay_coords_by_element(mol, conf,
    pdb_coords, res_name):
    heavy_atoms = [a for a in mol.
        GetAtoms() if a.GetAtomicNum() != 1]
    # Priority matching by structural
    # identifiers (N, CA, C, O, CB)
    for i, atom in enumerate(heavy_atoms):
        if i < len(pdb_order):
            name = pdb_order[i]
            if name in pdb_coords:
                coord = pdb_coords[name]
                conf.SetAtomPosition(
                    atom.GetIdx(), (coord[0], coord[1],
                        coord[2]))
```

This ensures that the RDKit Mol object possesses both the correct chemical graph (bonds, valences, aromaticity) and the accurate spatial distribution necessary for subsequent interaction mapping. A geometric centroid of the pocket is also calculated based on the Alpha-Carbon (C_α)

coordinates to allow for distance-based sorting and spatial orientation.

2.2. Feature Extraction and Pharmacophore Assignment

Once the molecule is accurately instantiated in 3D space, the algorithm extracts functional pharmacophore features that define protein-ligand interactions. We utilize RDKit’s Feature Factory (`BaseFeatures.fdef`) to identify distinct chemical behaviors associated with specific atomic clusters. The features tracked include:

- **HBondDonor** and **HBondAcceptor**
- **Hydrophobe**
- **Aromatic**
- **PosIonizable** and **NegIonizable**

For each identified feature, its 3D spatial coordinate is calculated as the centroid of its participating atoms. To evaluate the relative strength or "intensity" of these features, we apply specific metrics. For hydrophobicity, we utilize atomic Crippen LogP contributions. For hydrogen bonds, an empirical pseudo-intensity calculation provides continuous interaction strengths simulating partial charges and donor/acceptor proximity potentials.

Listing 2: Extraction of Hydrophobic Features using Crippen Contributions

```
contribs = rdMolDescriptors.
    _CalcCrippenContribs(mol_no_h)
for atom_idx, (logp, _mr) in enumerate(
    contribs):
    if logp > 0.1: # Threshold for
        hydrophobicity
        coord = positions[atom_idx]
        features.append(AtomFeature(
            feature_type="Hydrophobe",
            coords=coord,
            atom_indices=[atom_idx],
            intensity=float(logp),
        ))
```

2.3. Site Selection and Topological Ordering

A single amino acid may contain multiple interaction features. To represent these efficiently on the quantum lattice, we must select an optimal number of representative sites, denoted as K . The software provides multiple strategies:

- **heavy_atoms**: K equals the heavy atom count.
- **active_features**: Determines K by counting the number of distinct pharmacophore

features extracted, merging those that are spatially adjacent within a 1.5 Å threshold.

- **groups:** An automated clustering strategy utilizing K-Means, dynamically selecting K via a penalized inertia score.
- **fixed:** Enforces a strict scalar threshold for uniform segment generation.

Once K is determined, we apply K-Means clustering to the 3D coordinates of the extracted features. The centroid of each cluster becomes a "representative interaction site," assigned the dominant pharmacophore type of its constituents and an intensity equal to their sum.

2.3.1 Covalent Topology Maintenance

A critical innovation in our pipeline is the maintenance of chemical connectivity. Simple spatial clustering disrupts the sequence of the polypeptide chain, which is devastating for sequential quantum evaluation. We restore this by mapping each of the K sites to the nearest heavy atom in the molecular graph. We then perform a Breadth-First Search (BFS) starting from the N-terminal backbone atom.

Listing 3: BFS Topological Reordering

```
# BFS from the representative atom with
# the lowest index
start = min(site_to_atom)
visited = {} # atom_idx -> visit order
queue = deque([start])
order = 0
seen = set([start])
while queue:
    node = queue.popleft()
    visited[node] = order
    order += 1
    for nb in sorted(adj[node]):
        if nb not in seen:
            seen.add(nb)
            queue.append(nb)
```

The sites are ordered based on the sequence in which their associated atoms are discovered during the BFS traversal. This topological ordering ensures that the resulting 1D sequence of sites mathematically reflects the continuous chain of the physical protein.

2.4. Dimensionality Reduction: 3D to 2D Lattice Fitting

Quantum algorithms simulating physical grids currently scale more effectively in lower dimensions due to qubit constraints. Therefore, the

ordered 3D interaction sites must be mapped to a 2D plane. The pipeline supports two continuous projection techniques:

1. **Principal Component Analysis (PCA):** Maximizes the variance of the projected coordinates, providing a fast and deterministic linear transformation that aligns the interaction sites along their most elongated axes.
2. **Multi-Dimensional Scaling (MDS):** A non-linear technique that explicitly minimizes the stress function, preserving the pairwise Euclidean distances between the 3D sites as closely as possible in 2D. This is particularly advantageous for highly globular pockets.

The continuous 2D coordinates are subsequently normalized by dividing by a predefined `lattice_spacing` (e.g., 1.5 Å), effectively converting distances into unitless lattice steps.

2.5. Integer Snapping and Optimal Assignment

The continuous coordinates must be snapped to discrete integer grid nodes (i, j) to form the final lattice. A naive rounding approach (`round`) inevitably results in collisions, where multiple sites collapse onto the exact same grid node, losing critical interaction data.

To prevent this, we model the snapping problem as a classic assignment optimization task, solved via the Hungarian algorithm (Linear Sum Assignment). We dynamically generate a sufficiently large set of candidate integer grid nodes around the projected continuous coordinates. We define a cost matrix C where $C_{m,n}$ represents the squared Euclidean distance between the continuous site m and the candidate integer node n :

$$C_{m,n} = (x_m - i_n)^2 + (y_m - j_n)^2$$

Listing 4: Cost Matrix Generation for Hungarian Snapping

```
cost = np.zeros((k, n_cand))
for i, (x, y) in enumerate(coords_2d):
    for j, (ci, cj) in enumerate(candidates):
        cost[i, j] = (x - ci) ** 2 + (y - cj) ** 2
row_ind, col_ind = linear_sum_assignment(cost)
```

The Hungarian algorithm efficiently finds a one-to-one bipartite matching that assigns every site

to a unique grid node while minimizing the global sum of squared displacements. This guarantees a collision-free discrete lattice that optimally preserves the original spatial geometry.

2.6. Label Encoding

The final classical step is translating the categorical pharmacophore labels into numeric tensors. Depending on the downstream machine learning or quantum interface, labels are encoded as:

- **One-hot vectors:** Binary arrays separating feature types.
- **Index arrays:** Integer vocabulary mapping.
- **Dense embeddings:** Continuous vector representations.

3. Quantum Algorithm Implementation

With the binding pocket processed into a discrete chain of interaction properties, we transition to the quantum search domain. The quantum pipeline, implemented in `qubit_mapping.py`, utilizes IBM’s Qiskit SDK to execute a hybrid Grover’s Search and SWAP test architecture. We model two primary interaction forces for this phase: Hydrophobic (H) and Hydrogen-bond (B) potentials.

3.1. Qubit Mapping and Binary Encoding

Each discrete interaction site on the lattice is allocated 2 qubits: one for the hydrophobic property (q_H) and one for the H-bond property (q_B). To execute the initial coarse-grained search, we employ a Binary Threshold Mapping (Encoding 1). Given a bounded range of possible values for hydrophobicity $[h_{min}, h_{max}]$ and H-bonds $[hb_{min}, hb_{max}]$, we assign qubit states based on a midpoint threshold:

$$|q_H\rangle = \begin{cases} |0\rangle & \text{if } h < \frac{h_{min} + h_{max}}{2} \\ |1\rangle & \text{otherwise} \end{cases}$$

$$|q_B\rangle = \begin{cases} |0\rangle & \text{if } hb < \frac{hb_{min} + hb_{max}}{2} \\ |1\rangle & \text{otherwise} \end{cases}$$

The state of a single site is thus $|q\rangle = |q_H\rangle \otimes |q_B\rangle$. A ligand composed of L sites is concatenated into a target quantum state vector $|q_L\rangle$ of length $2L$ qubits.

3.2. Protein Superposition State Generation

The target protein pocket contains M sites, where $M > L$. The algorithm slides a window of length L across the protein sequence, generating a set of overlapping sub-segments. Duplicate windows are filtered out, leaving N unique binary segments.

We initialize a uniform quantum superposition state encompassing these N unique windows. By utilizing Qiskit’s `StatePreparation` library, we construct the state $|s\rangle$:

$$|s\rangle = \frac{1}{\sqrt{N}} \sum_{i=1}^N |x_i\rangle$$

where $|x_i\rangle$ represents the $2L$ -qubit basis state of the i -th unique protein window. This superposition allows the quantum computer to evaluate all possible docking sites simultaneously.

3.3. Modified Grover’s Search

To locate the protein window that matches the binary signature of the target ligand, we construct a modified Grover’s search circuit. The standard Grover iteration consists of two main unitary operators applied iteratively to the superposition state $|s\rangle$:

1. The Phase Oracle (O): The oracle identifies the target ligand state $|x_{target}\rangle$ and flips its phase, acting as an identity matrix with a -1 on the diagonal corresponding to the target index:

$$O = I - 2|x_{target}\rangle\langle x_{target}|$$

2. The Diffusion Operator (G): The diffusion operator (or inversion about the mean) amplifies the probability amplitude of the marked state while suppressing the others:

$$G = 2|s\rangle\langle s| - I$$

The number of optimal Grover iterations required to maximize the probability of measuring the target state is mathematically defined as:

$$\text{Iterations} = \left\lceil \frac{\pi}{4} \sqrt{N} \right\rceil$$

Following the application of the Grover iterations, the qubits are measured. Due to quantum shot noise and the discrete nature of the binary thresholding, multiple protein windows may exceed the threshold probability $\frac{1}{N}$. These windows form our subset of "candidate docking sites."

3.4. Continuous Amplitude Encoding

While the binary Grover search is exceptionally fast, it lacks the precision needed to rank the candidate sites effectively, as slight variations in interaction strengths are lost in thresholding. To resolve this, we map the candidates and the ligand into a continuous probability space using Amplitude Encoding (Encoding 2).

For each continuous property value (e.g., h), we calculate normalized probability amplitudes a and b such that $|a|^2 + |b|^2 = 1$:

$$a = \frac{|h_{max} - h|}{\sqrt{|h_{max} - h|^2 + |h - h_{min}|^2}}$$

$$b = \frac{|h - h_{min}|}{\sqrt{|h_{max} - h|^2 + |h - h_{min}|^2}}$$

The state of the hydrophobic qubit is then initialized as $|q_H\rangle = a|0\rangle + b|1\rangle$. An identical formulation computes amplitudes c and d for the H-bond qubit.

The full state vector for a single site is the tensor product:

$$|q_{site}\rangle = ac|00\rangle + ad|01\rangle + bc|10\rangle + bd|11\rangle$$

This encoding allows the entire spectrum of interaction affinities to be represented faithfully in the quantum amplitudes.

3.5. Quantum Euclidean Distance via SWAP Test

To determine which of the candidate docking sites is the absolute best match for the ligand, we calculate the quantum Euclidean distance between the ligand’s amplitude-encoded statevector ($|L\rangle$) and each candidate’s statevector ($|C_i\rangle$). This is achieved using a quantum SWAP test circuit. The circuit requires an auxiliary control qubit initialized to $|0\rangle$, and two data registers holding $|L\rangle$ and $|C_i\rangle$.

1. A Hadamard gate (H) is applied to the control qubit, placing it in superposition $\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$.
2. Controlled-SWAP (CSWAP or Fredkin) gates are applied across all corresponding qubits in the data registers, utilizing the control qubit.
3. A second Hadamard gate is applied to the control qubit.

The probability of measuring the control qubit in the state $|0\rangle$ is directly related to the inner product (fidelity) of the two quantum states:

$$P(ctrl = 0) = \frac{1}{2} + \frac{1}{2} |\langle L|C_i\rangle|^2$$

We extract the final Euclidean distance D using the normalization factor $Z = ||L||^2 + ||C_i||^2$:

$$D = \sqrt{4Z \cdot (P(ctrl = 0) - 0.5)}$$

By executing the SWAP test across all candidates identified by Grover’s search, Q-MODE rigorously ranks the sites, identifying the global minimum Euclidean distance as the optimal protein-ligand docking position.

4. System Performance and Experimental Setup

The classical processing component is optimized for typical academic workstations. Reading a pocket constraint from standard sources (e.g., the `1a08_pocket.pdb` sample) requires mere seconds. The computational bottleneck in classical chemoinformatics primarily lies in the RD-Kit conformer generation, which scales with the size of the molecule.

The quantum simulation using IBM AerSimulator scales exponentially in memory if simulating standard statevectors. The 2-qubit-per-site map limits standard local simulations to ligands with roughly 10-12 sites (requiring 20-24 qubits) before RAM limitations present a ceiling for desktop environments. The SWAP test requires $2n+1$ qubits (where n is the state length), pushing the requirement to 41-49 qubits, firmly positioning this application for execution on real NISQ hardware architectures.

5. Discussion and Algorithmic Complexity

The classical chemoinformatics pipeline significantly reduces the complexity of 3D molecular modeling. By employing MDS and the Hungarian algorithm, the projection from $\mathbb{R}^3$ to an $\mathbb{Z}^2$ grid preserves essential steric constraints without the exponential overhead of voxel-based convolution or heavily parameterized force fields.

On the quantum side, the separation of concerns into a binary Grover search followed by an amplitude-encoded SWAP test is crucial for near-term implementation. A pure amplitude-encoded Grover search would require an exponentially deep circuit to build a complex phase oracle capable of recognizing non-orthogonal states. By using the binary threshold (Encoding 1), the oracle matrix remains diagonal

with boolean properties, achieving the theoretical $O(\sqrt{N})$ quantum speedup over classical $O(N)$ linear scans. The SWAP test requires a circuit depth that scales linearly with the number of qubits $O(2L)$, but is only applied to a drastically reduced subset of candidates.

6. Conclusion

The Q-MODE project successfully engineers a complete, automated pipeline mapping real-world biochemical structures to quantum computational arrays. By bridging robust RD-Kit feature extraction and topology management with Qiskit’s quantum state simulators, we demonstrate a viable architecture for quantum-accelerated drug discovery. The classical inner-lattice mapping module elegantly resolves the theoretical assumptions of current quantum docking algorithms, while the dual-encoded quantum search provides both the speedup of Grover’s iteration and the precision of the SWAP test. Future developments will focus on adapting the pipeline for Noisy Intermediate-Scale Quantum (NISQ) hardware deployment and expanding the lattice dimensionality, mitigating current qubit connectivity limitations.